KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 30 · FINAL CHECKPOINT

Final Checkpoint — Voices of Change Review — Teacher Lesson Plan & Answer Key

Learning objectives: Students show mastery of Unit 6 (Weeks 26–29): numerical expressions and the order of operations, powers of 10 and exponents, multiplying and dividing decimals, and multiplying fractions as scaling and for area, plus myths and allusions, mysteries (clues, red herrings and inference), reading several sources to answer one question, and persuasive writing and evaluating arguments. Students also solve multi-step problems that mix weeks, judge explanations, find mistakes, edit sentences, and plan a short persuasive paragraph. This is the last lesson of KCE Pathways.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · scrap paper · printed Final Checkpoint Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: the four Unit 6 math settings (Kavya's robot rally in Detroit, the Motor City; the Adler Planetarium in Chicago; Lavinia's cross-country meet in Boulder, Colorado; Liora's tiny-house workshop in Austin, Texas) and the reading texts about the Prometheus statue at Rockefeller Center, two Savannah, Georgia, mysteries, the monarch butterflies of Michoacán, Mexico, and a skate park debate in Flagstaff, Arizona. The student types first and last name so results are sent. Read the scoring rules together: 2 tries per question, Pathway Points only for a correct first try, the answer plus the reason appear after the 2nd miss, and every section has a week tag so a missed item points to the week to review.
2. Slide 1 — Pencil Break (Pre): the student opens or prints the Final Checkpoint Pre/Post sheet (grade4-5/prepost/KCE_Checkpoint_PrePost_G4-5W30.pdf) and completes Page 1 (Final Checkpoint Warm-Up) off screen. Do not help with answers yet; it shows where the student is starting.
3. Slide 2 — Week 26 Math: Order of Operations, Part 1 (5 pts: 3 + 2), with Zuberi's practice rally in Detroit and Ms. Sorensen, the judge. Each skill slide has a one-line reminder at the top; if a student misses an item, point back to it.
4. Slide 3 — Week 26 Math: Writing Expressions and Parentheses, Part 2 (2 pts): write and evaluate an expression, place one pair of parentheses, or find the change from $100.
5. Slide 4 — Week 26 Reading: Myths and Allusions, Part 1 (5 pts: 3 + 2), with a short original story in each set ("The Science Fair" or "The Pet Vote," with Cordelia and her cousin Rafferty). Remind: an allusion is never meant literally.
6. Slide 5 — Week 26 Reading: Dialogue, Compound Sentences, and Its, Part 2 (2 pts): a comma before a speaker tag, a comma before but, or its; checked exactly.
7. Slide 6 — Week 27 Math: Powers of 10 and Exponents, Part 1 (5 pts: 3 + 2), with Ifeoma at the Adler Planetarium and Dr. Fennimore, an astronomy educator. Tip: the exponent counts the factors of 10.
8. Slide 7 — Week 27 Math: Space Numbers, Part 2 (2 pts): the Moon's distance as 2.39×10^{n}, $64 \div 10^{3}$, or how far light travels in 10^{3} seconds.

9. Slide 8 — Week 27 Reading: Clues, Red Herrings, and Inference, Part 1 (5 pts: 3 + 2), with a short new mystery in each set ("The Missing Cash Box," "The Banana Mystery," or "The Fallen Sunflower"). Ask: does EVERY clue fit the answer?
10. Slide 9 — Week 27 Reading: Dialogue, Titles, and Subject-Verb Agreement, Part 2 (2 pts): checked exactly (the verb item accepts either verb tense).
11. Slide 10 — Week 28 Math: Multiplying and Dividing Decimals, Part 1 (5 pts: 3 + 2), with Aurelio's race week in Boulder and Coach Brisa. Tip: predict first; times a number less than 1 gets smaller, divided by a number less than 1 gets bigger.
12. Slide 11 — Week 28 Math: Decimals on the Course, Part 2 (2 pts): a product, a quotient, or a map-scale distance.
13. Slide 12 — Week 28 Reading: Reading Several Sources, Part 1 (5 pts: 3 + 2), with Saoirse and Emeka's note cards from Sources A, B and C in each set. Ask which source said what.
14. Slide 13 — Week 28 Reading: Report Writing, Part 2 (2 pts): fewer or less, than or then, or a period inside the closing quotation mark; checked exactly.
15. Slide 14 — Week 29 Math: Scaling and Area with Fractions, Part 1 (5 pts: 3 + 2), with Xiomara's tiny-house plans in Austin and Mr. Ulrich, the carpenter. Tip: compare the factor with 1 before multiplying.
16. Slide 15 — Week 29 Math: Fractions in the Workshop, Part 2 (2 pts): 1 1/4 × 20, the area of a cutting board as a fraction, or a main floor plus a loft 2/7 as large.
17. Slide 16 — Week 29 Reading: Persuasive Writing and Arguments, Part 1 (5 pts: 3 + 2), with Valentina's letter to the Library Board, Mr. Percival's speech, or a summary of the meeting in each set (made up, with made-up numbers).
18. Slide 17 — Week 29 Reading: Letter Writing, Part 2 (2 pts): a letter greeting (comma or colon accepted), could have, or commas around a state; checked exactly.
19. Slide 18 — Putting Skills Together, Part 1 (6 pts, 3 each): a multi-step problem that mixes two weeks and a "which explanation is correct?" item.
20. Slide 19 — Be the Checker, Part 2 (6 pts, 3 each): a math "find the mistake" (type the correct answer) and a reading "find the mistake" (choose the explanation). Ask the student to say why each wrong choice is wrong.
21. Slide 20 — Fix the Sentence, Part 1 (4 pts, 2 each): typed sentences checked exactly (quotation marks and end marks in dialogue, their, you're, its). Straight or curly quotes and extra spaces are OK.
22. Slide 21 — Fix the Sentence, Part 2 (4 pts, 2 each): a story title, commas in a place, a date or a renaming phrase, and find the real subject before choosing the verb; each verb item accepts either verb tense.
23. Slide 22 — Writer's Notebook: Make Your Case: a persuasive paragraph of 3 to 5 sentences to the family about which Unit 6 place to visit first (practice, XP only), then one scored follow-up item (4 pts) about a transition, a quotation, or a claim with because. Read the notebook writing aloud together if time allows.
24. Slide 23 — Match-Up Round (10 pts): 5 Unit 6 vocabulary words across math and reading, 2 pts each on the first try.
25. Slide 24 — Looking Back on Unit 6 reflection (bonus XP only).
26. Slide 25 — Unit 6 Boss Challenge (10 pts, 5 each): a three-step math problem with a different situation in each set (Orsolya's robotics parts in Detroit, Augustin's garden bed in Austin, or Ifeoma's scale model at the Adler Planetarium) and a reading/writing item that combines two skills (a reasoning trap, a red herring and an allusion, or a synthesis with citations).
27. Slide 26 — Pencil Break (Post): the student completes Page 2 (Final Checkpoint Challenge) of the same sheet (grade4-5/prepost/KCE_Checkpoint_PrePost_G4-5W30.pdf), plus the ★ Challenge if time allows, before seeing the score.
28. Slide 27 — Results (Unit 6 Complete!): the score out of 100 is sent to KingCretot Experience, with a score breakdown by week so you can see what to review. 85 or higher shows the "KCE Pathways Graduate!" badge, "You Graduated from KCE Pathways!" with the message "Congratulations, graduate! Unit 6 and all of KCE Pathways are complete!", a big confetti burst, and a "Print your certificate" button; below 85 shows "Good Work on the Final Checkpoint!" with encouragement ("You are getting closer every time."), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Final Checkpoint Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

ORDER OF OPERATIONS (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
a1_s0	Set 1: *Zuberi's Practice Rally Zuberi drives his team's robot at a practice rally in Detroit, Michigan, the Motor City. Ms. Sorensen, the judge, writes every score as a numerical expression.* Two steps: Zuberi's robot stacks 7 cubes worth 4 points each, parks for 12 points, and then loses 5 points for a penalty. Ms. Sorensen writes 7 × 4 + 12 − 5. What is the score? Points:	**35** (also accepted: 35 points, 35 point, 35 pts)	Multiply first: 7 × 4 = 28. Then add and subtract from left to right: 28 + 12 = 40, and 40 − 5 = 35 points.
a2_s0	Set 1: What is the value of 40 − 8 × 3 + 2? [98 / 18 / 14 / 0]	**18**	Multiply first: 8 × 3 = 24. Then left to right: 40 − 24 = 16, and 16 + 2 = 18. (98 works left to right without multiplying first, and 14 adds 24 + 2 before subtracting.)
a1_s1	Set 2: *Zuberi's Practice Rally Zuberi drives his team's robot at a practice rally in Detroit, Michigan. Ms. Sorensen, the judge, checks each team's plan before she writes the score.* Two steps: Zuberi's robot earns 15 points in each of 3 rounds, but Ms. Sorensen takes away a 6-point penalty in EVERY round. She writes 3 × (15 − 6). What is the score? Points:	**27** (also accepted: 27 points, 27 point, 27 pts)	Parentheses first: 15 − 6 = 9 points for each round. Then multiply: 3 × 9 = 27 points.
a2_s1	Set 2: Which expression means "subtract 8 from 50, then divide by 6"? [50 − 8 ÷ 6 / (8 − 50) ÷ 6 / (50 − 8) ÷ 6 / 6 ÷ (50 − 8)]	**(50 − 8) ÷ 6**	Subtract 8 from 50 means start with 50: 50 − 8. The word then means that step comes first, so it goes in parentheses before dividing by 6.
a1_s2	Set 3: *Zuberi's Practice Rally Zuberi drives his team's robot at a practice rally in Detroit, Michigan. Ms. Sorensen writes some scores with brackets and parentheses.* Two steps: work from the inside out. Ms. Sorensen's score sheet says 5 × [18 − (4 + 6)]. What is the score? Points:	**40** (also accepted: 40 points, 40 point, 40 pts)	Inside the parentheses: 4 + 6 = 10. Inside the brackets: 18 − 10 = 8. Last step: 5 × 8 = 40.
a2_s2	Set 3: Without computing: 6 × (708 + 59) is how many times as large as 2 × (708 + 59)? [4 times / 12 times / 6 times / 3 times]	**3 times**	Both expressions multiply the same group, 708 + 59. 6 is 3 times as large as 2, so the first expression is 3 times as large. No adding needed!

WRITING EXPRESSIONS AND PARENTHESES (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
a3_s0	Set 1: Two steps: write an expression for "add 9 and 15, then multiply by 4," and evaluate it. What is the value? Value:	**96**	The words say add first, so the expression is (9 + 15) × 4. Parentheses: 9 + 15 = 24. Then 24 × 4 = 96.
a3_s1	Set 2: Two steps: add ONE pair of parentheses so that 3 + 5 × 4 − 2 = 30. Test your choice, then type the whole left side. Keep the numbers in the same order. (Use x or * for times if you need to.) Left side:	**(3+5)×4−2** (also accepted: (3+5)×4-2, (3+5)x4−2, (3+5)x4-2, (3+5)*4−2, (3+5)*4-2)	Try (3 + 5) × 4 − 2: 3 + 5 = 8, then 8 × 4 = 32, and 32 − 2 = 30. Other placements give 13 or 21.
a3_s2	Set 3: Two steps: Zuberi buys 5 battery packs that cost $18 each and pays with a $100 bill. His change is 100 − 5 × 18. How many dollars of change does he get? Dollars:	**10** (also accepted: 10.00, 10 dollars, 10 dollar, 10.00 dollars, 10.00 dollar)	Multiply first: 5 × 18 = 90. Then subtract: 100 − 90 = $10.

MYTHS AND ALLUSIONS (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: *From "The Science Fair," an original story 1Cordelia had spent three whole weekends building a model volcano for the science fair. "Getting the clay to stand up was a Herculean job," she told her cousin Rafferty. 2Rafferty had made his poster in one evening, and the judges loved it. "Of course," Cordelia sighed. "You have the Midas touch." But when her volcano finally erupted in a fizz of red foam, the whole crowd cheered.* Two steps: Step 1, which myth does "You have the Midas touch" point to? Step 2, what does Cordelia mean? [King Midas; Rafferty's poster really turns into gold. / King Midas; everything Rafferty tries seems to succeed easily. / Achilles; Rafferty has one weak spot. / Pandora; Rafferty starts a lot of trouble.]	**King Midas; everything Rafferty tries seems to succeed easily.**	The phrase points to King Midas, whose touch turned everything to gold. As an allusion it is not literal: Cordelia means that everything Rafferty tries seems to succeed easily.
b2_s0	Set 1: Type your answer: in paragraph 1, "a Herculean job" is an allusion to a hero who had to complete twelve almost impossible tasks. Type the hero's name. Hero:	**hercules** (also accepted: heracles)	"Herculean" comes from Hercules, the Roman name for the Greek hero Heracles, who had to complete twelve almost impossible tasks. Cordelia means the job was huge and exhausting.

ID	Question	Answer	Why (explain to student)
b1_s1	Set 2: *From "The Science Fair," an original story 1Rafferty was the best speller in his class, but he could never remember how to spell necessary. "That word is my Achilles' heel," he admitted. 2His little sister, Matilda, followed him around the kitchen. "One c, two s's," he muttered. "Two s's!" Matilda repeated. "All right, Echo," Rafferty laughed. "That's enough."* Which explanation of "That word is my Achilles' heel" is correct? [Rafferty's heel hurts whenever he spells. / Rafferty spells every word perfectly, like a Greek hero. / Spelling that one word is the weak spot of a strong speller, the way Achilles had one unprotected spot, his heel. / Rafferty started something that caused many problems.]	**Spelling that one word is the weak spot of a strong speller, the way Achilles had one unprotected spot, his heel.**	An Achilles' heel is one weak spot in someone or something strong. In the later tellings, the heel his mother held was the only unprotected spot on Achilles. Rafferty is a strong speller with one weak word.
b2_s1	Set 2: Type your answer: two steps. Rafferty calls Matilda "Echo." In the myth, the goddess Juno made Echo able to repeat only the ___ words she heard. Type one word. Word:	**last** (also accepted: the last)	In the Roman poet Ovid's telling, Juno made Echo able to repeat only the last words she heard. Matilda keeps repeating the last words Rafferty says.
b1_s2	Set 3: *From "The Pet Vote," an original story 1On a class trip to Rockefeller Center in New York City, Cordelia stood below the gilded statue of Prometheus that the sculptor Paul Manship made in 1934. 2Back at school, Rafferty asked if the class could keep a pet lizard, and the teacher said yes. By Friday, everyone wanted a pet: a hamster, a snake, even a pony. "Uh-oh," the teacher groaned. "I think I just opened Pandora's box."* In paragraph 2, what does the teacher mean by "I think I just opened Pandora's box"? [She found a real box full of pets. / Pets are her one weak spot. / Everything she touches succeeds. / Saying yes to one pet started many problems she did not expect.]	**Saying yes to one pet started many problems she did not expect.**	Opening Pandora's box means starting something that leads to many problems no one expected. In the myth, troubles flew out when Pandora lifted the lid. Saying yes to one pet led to many more pet requests.
b2_s2	Set 3: Type your answer: two steps. In the Greek myth, the figure shown by the statue in paragraph 1 took something from the gods and gave it to humans. What did he take? Type one word. Word:	**fire**	In the Greek myth, Prometheus took fire from the gods and carried it to humans.

DIALOGUE, COMPOUND SENTENCES, AND ITS (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b3_s0	Set 1: Fix this sentence: "That word is my weak spot" Rafferty admitted.	**"That word is my weak spot," Rafferty admitted.**	When the speaker tag comes after the spoken words, a comma goes inside the closing quotation mark.
b3_s1	Set 2: Fix this sentence: Echo loved Narcissus but he turned her away.	**Echo loved Narcissus, but he turned her away.**	Echo loved Narcissus and he turned her away are both complete sentences. When but joins two complete sentences, put a comma before it.

ID	Question	Answer	Why (explain to student)
b3_s2	Set 3: Fix this sentence: The volcano spilled it's red foam across the table.	**The volcano spilled its red foam across the table.**	The foam belongs to the volcano, so use the possessive its, with no apostrophe. It's means it is.

POWERS OF 10 AND EXPONENTS (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
c1_s0	Set 1: *Ifeoma's Planetarium Puzzles Ifeoma volunteers at the Adler Planetarium in Chicago, Illinois, which opened in 1930 as the first planetarium in the Western Hemisphere. Dr. Fennimore, an astronomy educator, gives her number puzzles.* Two steps: the dome shows 4×10^4 stars. Then Dr. Fennimore adds 6×10^2 more stars. How many stars are on the dome in all? Write the number in standard form. Stars:	**40,600** (also accepted: 40,600 stars; 40,600 star)	$4 \times 10^4 = 4 \times 10{,}000 = 40{,}000$ and $6 \times 10^2 = 6 \times 100 = 600$. In all: 40,000 + 600 = 40,600 stars.
c2_s0	Set 1: Which has the GREATEST value? [9×10^4 / 10^5 / 199,999 / 2×10^5]	**2×10^5**	$2 \times 10^5 = 200{,}000$. The others: $9 \times 10^4 = 90{,}000$, $10^5 = 100{,}000$, and 199,999 is 1 less than 200,000.
c1_s1	Set 2: *Ifeoma's Planetarium Puzzles Ifeoma volunteers in the gift shop of the Adler Planetarium, on the shore of Lake Michigan in Chicago, Illinois. Dr. Fennimore asks her to check the prices.* Two steps: a box holds 10^3 star stickers, and each sticker costs $0.08. Ifeoma buys one box and pays with a $100 bill. How much change does she get? Dollars:	**20** (also accepted: 20.00, 20 dollars, 20 dollar, 20.00 dollars, 20.00 dollar)	A box holds $10^3 = 1{,}000$ stickers. Cost: $0.08 \times 1{,}000 = \$80$ (each digit moves 3 places to the left). Change: 100 − 80 = $20.
c2_s1	Set 2: In 7.3×10^3, how far does each digit move, and in which direction? [3 places to the left / 3 places to the right / 7 places to the left / 2 places to the left]	**3 places to the left**	Multiplying by 10^3 means three factors of 10, so every digit moves 3 places to the LEFT and becomes larger: $7.3 \times 10^3 = 7{,}300$.
c1_s2	Set 3: *Ifeoma's Planetarium Puzzles Ifeoma helps Dr. Fennimore get ready for a school group at the Adler Planetarium in Chicago, Illinois.* Two steps: Dr. Fennimore has 5,600 grams of modeling clay for planet models. He shares it equally among 10^2 groups. Then each group shares its clay equally among 10 students. How many grams does each student get? Grams:	**5.6** (also accepted: 5.6 grams, 5.6 gram, 5.6 g)	Each group: $5{,}600 \div 10^2 = 5{,}600 \div 100 = 56$ grams. Each student: $56 \div 10 = 5.6$ grams.
c2_s2	Set 3: $27 \div 10^n = 0.0027$. What is n? [3 / 4 / 2 / 5]	**4**	The 7 in 27 moved from the ones place to the ten-thousandths place, which is 4 places to the right, so n = 4. Check: $27 \div 10{,}000 = 0.0027$.

SPACE NUMBERS (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
c3_s0	Set 1: Two steps: the Moon is about 239,000 miles from Earth on average. Ifeoma writes this distance as 2.39×10^{n} miles. What number is n? n =	**5** (also accepted: five)	2.39×10^{5}: each digit moves 5 places to the left: 2.39 → 23.9 → 239 → 2,390 → 23,900 → 239,000. So n = 5.
c3_s1	Set 2: Two steps: write 10^{3} in standard form, then divide 64 by it. What is the quotient? Quotient:	**0.064**	10^{3} = 1,000. Dividing by 1,000 moves each digit 3 places to the right: 64 → 6.4 → 0.64 → 0.064. A placeholder zero holds the tenths place.
c3_s2	Set 3: Two steps: light travels about 186,000 miles in one second. About how many miles does light travel in 10^{3} seconds? Miles:	**186,000,000** (also accepted: 186 million; 186,000,000 miles; 186,000,000 mile; 186 million miles; 186 million mile)	10^{3} seconds = 1,000 seconds. 186,000 × 1,000 moves every digit 3 places to the left: 186,000,000 miles.

CLUES, RED HERRINGS, AND INFERENCE (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
d1_s0	Set 1: *From "The Missing Cash Box," an original mystery 1When Octavia arrived at the library book sale, the cash box was missing from the table. Her friend Lucius pointed at a boy in a muddy soccer uniform. "He ran right past the table a minute ago!" 2Octavia noticed that the boy was carrying his own water bottle and nothing else. Then she spotted a key on a ribbon hanging from the librarian's desk drawer, and the drawer was not quite closed. Inside sat the cash box. Mr. Abernathy, the librarian, had locked it away before his lunch break.* Two steps: which detail is the RED HERRING, and why? [The drawer that was not quite closed: it hid the truth. / The key on a ribbon: keys are never real clues. / The cash box in the drawer: it was never missing. / The boy running past the table: he was carrying only his own water bottle.]	**The boy running past the table: he was carrying only his own water bottle.**	The running boy is the loud clue that makes him look guilty, but he carried only his water bottle. The quiet clues, the key on the ribbon and the drawer that was not quite closed, lead to the truth.
d2_s0	Set 1: Type your answer: who had locked the cash box away in the drawer? Type his name. Name:	**mr. abernathy** (also accepted: mr abernathy, abernathy, mister abernathy, the librarian, librarian)	Paragraph 2: "Mr. Abernathy, the librarian, had locked it away before his lunch break."

ID	Question	Answer	Why (explain to student)
d1_s1	Set 2: *From "The Banana Mystery," an original mystery 1I was sure my brother Fitzroy had eaten the last banana. There was a peel in the kitchen trash, and he was grinning. 2Then Mom said Fitzroy had been at swim practice since noon, and his coach had driven him home just five minutes earlier. I looked again: the peel was brown and dry, as if it had been there for days. The last banana was in my own backpack, squashed under my math book.* Two steps: who is telling this mystery, and how do you know? [A character in the story, in the first person, because the teller uses I and my. / A narrator outside the story, because the story names Fitzroy. / Fitzroy, because he is grinning. / Mom, because she explains where Fitzroy was.]	**A character in the story, in the first person, because the teller uses I and my.**	The narrator uses I, my, and my own, so a character inside the story is telling it in the first person. Fitzroy is called "my brother," so he is not the teller.
d2_s1	Set 2: Type your answer: Fitzroy was at swim practice, and his coach drove him home. In a mystery, proof like this that a suspect could not have done it is called an ___. Word:	**alibi** (also accepted with "a" or "an" or in the plural)	Proof that a suspect was somewhere else, or could not have done it, is an alibi. Fitzroy was at swim practice, and his coach can say so.
d1_s2	Set 3: *From "The Fallen Sunflower," an original mystery 1Someone had knocked over the class sunflower. Hortense blamed Pipkin, the class rabbit, because there was a tuft of white fur on the windowsill. 2But Pipkin's cage was still locked, and the teacher had kept the only key in her pocket all morning. The fur matched the fuzzy white sweater Hortense herself was wearing. Then a gust of wind blew through the open window and slammed it against the flowerpot as Hortense watched.* Which inference is BEST supported by ALL the clues? [Pipkin the rabbit knocked over the sunflower. / The wind blew the open window into the flowerpot. / The teacher knocked it over to test the class. / Someone must have stolen the key from the teacher.]	**The wind blew the open window into the flowerpot.**	Every clue fits the wind: the cage was locked, so Pipkin could not get out; the fur came from Hortense's sweater; and the wind slammed the open window against the pot. A strong inference uses ALL the clues.
d2_s2	Set 3: Type your answer: a story's message about life, such as "Look at all the evidence before you blame someone," is called its ___. Word:	**theme** (also accepted: the theme)	The lesson about life that a story teaches is its theme. Both of the Week 27 mysteries taught this same theme.

DIALOGUE, TITLES, AND SUBJECT-VERB AGREEMENT (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
d3_s0	Set 1: Fix this sentence: "He ran past the table" Lucius said.	**"He ran past the table," Lucius said.**	When the speaker tag comes after the spoken words, a comma goes inside the closing quotation mark.
d3_s1	Set 2: Fix this sentence (put the story's title in quotation marks): In The Missing Cash Box Octavia solves the case.	**In "The Missing Cash Box," Octavia solves the case.**	The title of a story goes inside quotation marks, and the comma after the title goes inside the closing quotation mark.

ID	Question	Answer	Why (explain to student)
d3_s2	Set 3: Fix this sentence: The clues in the classroom points to the wind.	**The clues in the classroom point to the wind.** (also accepted: The clues in the classroom pointed to the wind.)	The subject is clues (plural), not classroom, so the verb is point, not points. The words "in the classroom" do not change the subject.

MULTIPLYING AND DIVIDING DECIMALS (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: *Aurelio's Race Week Aurelio runs cross-country in Boulder, Colorado, a city 5,430 feet above sea level at the foot of the Flatirons. Coach Brisa records every time and pace.* Two steps: Aurelio finishes a 3-kilometer race in 14.7 minutes. If he kept that same pace, how many minutes would a 5-kilometer race take him? Minutes:	**24.5** (also accepted: 24 1/2, 24.5 minutes, 24.5 minute, 24.5 min, 24 1/2 minutes, 24 1/2 minute, 24 1/2 min)	Pace: 14.7 ÷ 3 = 4.9 minutes per kilometer. Five kilometers: 4.9 × 5 = 24.5 minutes.
e2_s0	Set 1: Which product equals 0.18? [0.3 × 6 / 0.3 × 0.6 / 0.9 × 0.9 / 0.3 + 0.6]	**0.3 × 0.6**	3 tenths × 6 tenths = 18 hundredths = 0.18. The others: 0.3 × 6 = 1.8, 0.9 × 0.9 = 0.81, and 0.3 + 0.6 = 0.9.
e1_s1	Set 2: *Aurelio's Race Week Aurelio's team hosts a cross-country meet in Boulder, Colorado. Coach Brisa runs the water table.* Two steps: Coach Brisa has 6.3 liters of water. She fills cups that hold 0.35 liter each. If 15 runners each take one full cup, how many full cups are left? Cups:	**3** (also accepted: three, 3 cups, 3 cup, three cup; with "a" or "an" or in the plural)	Cups: 6.3 ÷ 0.35. Multiply both by 100: 630 ÷ 35 = 18 cups. Left: 18 − 15 = 3 cups.
e2_s1	Set 2: To find 0.56 ÷ 0.07, Aurelio multiplies both numbers by the same power of 10. Which one makes the divisor a whole number? [10 / 7 / 100 / 0.1]	**100**	0.07 has two decimal places, so multiply both numbers by 100: 56 ÷ 7 = 8. Multiplying by 10 gives 5.6 ÷ 0.7, and the divisor is still a decimal.
e1_s2	Set 3: *Aurelio's Race Week Aurelio's team gets ready for a meet in Boulder, Colorado, where the Flatirons rise just west of town.* Two steps: Aurelio buys 2.5 pounds of trail mix at $3.60 per pound. He pays with a $20 bill. How much change does he get? Dollars:	**11** (also accepted: 11.00, 11 dollars, 11 dollar, 11.00 dollars, 11.00 dollar)	Trail mix: 2.5 × 3.60. Multiply 25 × 36 = 900, with two decimal places in all: $9.00. Change: 20 − 9 = $11.
e2_s2	Set 3: Without computing, which answer is GREATER than 9? [9 × 0.75 / 9 ÷ 1.5 / 9 × 0.5 / 9 ÷ 0.75]	**9 ÷ 0.75**	Dividing by a number less than 1 asks how many small pieces fit, so the quotient is greater: 9 ÷ 0.75 = 12. The others are smaller than 9: 6.75, 6, and 4.5.

DECIMALS ON THE COURSE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e3_s0	Set 1: Two steps: estimate, then multiply. What is 2.6 × 1.4? Product:	**3.64**	Estimate: 3 × 1 = 3. Multiply 26 × 14 = 364. There are two decimal places in all, so the product is 3.64, close to the estimate.

ID	Question	Answer	Why (explain to student)
e3_s1	Set 2: Two steps: make the divisor a whole number, then divide. What is 7.2 ÷ 0.9? Quotient:	**8**	Multiply both numbers by 10: 72 ÷ 9 = 8. Check: 8 × 0.9 = 7.2. The divisor is less than 1, so the quotient is greater than 7.2.
e3_s2	Set 3: Two steps: on the course map, each centimeter stands for 0.6 kilometer, and the route is 4.5 centimeters long. Aurelio has already run 1.9 kilometers of the real route. How many kilometers are left? Kilometers:	**0.8** (also accepted: 0.8 kilometers, 0.8 kilometer, 0.8 km)	Real route: 4.5 × 0.6 = 2.7 kilometers. Still to go: 2.7 − 1.9 = 0.8 kilometer.

READING SEVERAL SOURCES (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
f1_s0	Set 1: *Saoirse and Emeka's note cards. Each card names its source. 1Source A (article): Eastern monarchs spend the winter, from about October to March, in oyamel fir forests in central Mexico. The forests give them a cool, steady climate. 2Source B (fact sheet): Monarchs that hatch in late summer go into diapause. They do not lay eggs; instead, they save their energy for the long flight south.* Which sentence best SYNTHESIZES Source A and Source B? [Monarchs spend the winter in oyamel fir forests (Source A). / Late-summer monarchs save their energy instead of laying eggs (Source B), so they can reach cool fir forests in Mexico for the winter (Source A). / Late-summer monarchs save their energy (Source A), and the forests are cool (Source B). / The forests give them a cool, steady climate.]	**Late-summer monarchs save their energy instead of laying eggs (Source B), so they can reach cool fir forests in Mexico for the winter (Source A).**	A synthesis sentence combines facts from two or more sources in new words and names each source correctly. One choice uses only one source, one swaps the citations, and one copies Source A word for word with no citation.
f2_s0	Set 1: Type your answer: according to Source A, the monarchs stay in the fir forests from about October to what month? Month:	**march**	Source A says the monarchs stay "from about October to March."
f1_s1	Set 2: *Saoirse and Emeka's note cards. Each card names its source. 1Source A (article): In 2008, the Monarch Butterfly Biosphere Reserve was named a World Heritage Site by UNESCO. 2Source C (survey data): Forest covered by monarch colonies: 1996 to 1997 (the most ever recorded), about 45 acres; 2023 to 2024, 2.22 acres; 2024 to 2025, 4.42 acres; 2025 to 2026, 7.24 acres.* Research question: "How are the monarchs doing today?" Which note card is RELEVANT and cites its source correctly? [The reserve was named a World Heritage Site in 2008 (Source C). / The colonies covered 7.24 acres in 2025 to 2026 (Source A). / The colonies covered 7.24 acres in 2025 to 2026, far below about 45 acres in 1996 to 1997 (Source C). / The reserve was named a World Heritage Site in 2008 (Source A).]	**The colonies covered 7.24 acres in 2025 to 2026, far below about 45 acres in 1996 to 1997 (Source C).**	The acres in Source C answer "How are the monarchs doing today?", and the card names Source C correctly. The World Heritage honor is real history, but it does not measure how the monarchs are doing.

ID	Question	Answer	Why (explain to student)
f2_s1	Set 2: Type your answer: two steps. Using Source C, how many MORE acres did the colonies cover in 2025 to 2026 than in 2023 to 2024? Acres:	**5.02** (also accepted: 5.02 acres, 5.02 acre)	Source C: 7.24 acres in 2025 to 2026 and 2.22 acres in 2023 to 2024. 7.24 − 2.22 = 5.02 acres.
f1_s2	Set 3: *Saoirse and Emeka's note cards. Each card names its source. 1Source B (fact sheet): Females lay their eggs only on milkweed, the one plant monarch caterpillars depend on. 2Source C (survey data): Threats to monarchs include the loss of milkweed and flowering plants in the United States, drought and extreme weather, and illegal logging in the reserve's forests.* Find the mistake! Emeka writes, "Monarch caterpillars depend on milkweed (Source C), and the loss of milkweed is a threat to monarchs (Source B)." Which explanation is correct? [Emeka is right; nothing needs to change. / Monarch caterpillars eat oyamel fir needles, not milkweed. / He should cite Source A for both facts. / The facts are right, but the citations are switched: the milkweed fact is in Source B, and the threat is in Source C.]	**The facts are right, but the citations are switched: the milkweed fact is in Source B, and the threat is in Source C.**	Both facts are true, but they are cited to the wrong sources. Milkweed as the caterpillars' one food plant is in Source B, and the list of threats is in Source C. A correct fact with the wrong source is still a citation mistake.
f2_s2	Set 3: Type your answer: a fact that is connected to your research question and helps answer it is ___. (one word) Word:	**relevant**	A relevant fact is connected to your research question and helps answer it. Facts that are interesting but do not help get crossed out.

REPORT WRITING (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
f3_s0	Set 1: Fix this sentence (fewer or less): The table shows less acres than in the 1990s.	**The table shows fewer acres than in the 1990s.**	Acres can be counted, so use fewer. Use less for amounts you cannot count, like less water.
f3_s1	Set 2: Fix this sentence (than or then): The colonies in 2025 to 2026 were bigger then the colonies before.	**The colonies in 2025 to 2026 were bigger than the colonies before.**	Use than to compare. Then tells when something happens.
f3_s2	Set 3: Fix this sentence: Source B says the monarchs "save their energy for the long flight south".	**Source B says the monarchs "save their energy for the long flight south."**	In American English, a period at the end of a quotation goes inside the closing quotation mark.

SCALING AND AREA WITH FRACTIONS (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
g1_s0	Set 1: *Xiomara's Tiny-House Plans Xiomara helps at a tiny-house workshop in Austin, Texas, the capital of Texas. Many builders describe a tiny house as a home of 400 square feet or less, not counting lofts. Mr. Ulrich, the carpenter, checks every measurement.* Two steps: a sleeping loft is 6 2/3 feet long and 4 1/2 feet wide. What is its area in square feet? Square feet:	**30** (also accepted: 30 square feet, 30 square foot, 30 sq ft)	6 2/3 = 20/3 and 4 1/2 = 9/2. Area: 20/3 × 9/2 = 180/6 = 30 square feet.
g2_s0	Set 1: Without multiplying, which product is GREATER than 18? [4/5 × 18 / 18/18 × 18 / 1/2 × 18 / 5/4 × 18]	**5/4 × 18**	5/4 is greater than 1, so 5/4 × 18 is greater than 18 (it is 22 1/2). 4/5 and 1/2 are less than 1, and 18/18 equals 1.
g1_s1	Set 2: *Xiomara's Tiny-House Plans Xiomara helps Mr. Ulrich, the carpenter, cut boards at a tiny-house workshop in Austin, Texas.* Two steps: Mr. Ulrich has a 12-foot board. He cuts off 1/3 of it for a shelf. Then he uses 3/4 of the piece that is left for a step. How long is the step? Feet:	**6** (also accepted: 6 feet, 6 foot, 6 ft)	Shelf: 1/3 × 12 = 4 feet, so 12 − 4 = 8 feet are left. Step: 3/4 × 8 = 6 feet.
g2_s1	Set 2: A tile is 1/4 foot by 1/3 foot. What is its area? [1/12 square foot / 2/7 square foot / 1/7 square foot / 7/12 square foot]	**1/12 square foot**	Area = length × width: 1/4 × 1/3 = 1/12 square foot. Twelve tiles this size (4 columns and 3 rows) cover one square foot.
g1_s2	Set 3: *Xiomara's Tiny-House Plans Xiomara frames windows with Mr. Ulrich at a tiny-house workshop in Austin, Texas.* Two steps: each window is 1 1/2 feet wide and 2 2/3 feet tall. Xiomara frames 3 matching windows. What is their total area in square feet? Square feet:	**12** (also accepted: 12 square feet, 12 square foot, 12 sq ft)	One window: 1 1/2 × 2 2/3 = 3/2 × 8/3 = 24/6 = 4 square feet. Three windows: 3 × 4 = 12 square feet.
g2_s2	Set 3: Which equation shows multiplying by a fraction equal to 1? [2/7 × 3/7 = 6/49 / 2/7 × 3/3 = 6/21 / 2/7 + 3/3 = 1 2/7 / 2/7 × 3 = 6/7]	**2/7 × 3/3 = 6/21**	3/3 = 1, so 2/7 × 3/3 = 6/21 is an equivalent fraction: 6/21 names the same amount as 2/7.

FRACTIONS IN THE WORKSHOP (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
g3_s0	Set 1: Two steps: find 1 1/4 × 20 by adding 20 + (1/4 × 20). What is the product? Product:	**25**	1/4 of 20 is 5, so 1 1/4 × 20 = 20 + 5 = 25. The factor is greater than 1, so the product is greater than 20.

ID	Question	Answer	Why (explain to student)
g3_s1	Set 2: Two steps: a cutting board is 2/3 foot long and 3/5 foot wide. What is its area in square feet? Type a fraction. Square feet:	**2/5** (also accepted: 6/15, 4/10, 0.4, 2/5 square feet, 2/5 square foot, 6/15 square feet, 6/15 square foot, 4/10 square feet, 4/10 square foot, 0.4 square feet, 0.4 square foot)	Area: 2/3 × 3/5 = 6/15, which simplifies to 2/5 square foot.
g3_s2	Set 3: Two steps: the main floor of a tiny house is 280 square feet. The loft is 2/7 as large as the main floor. How many square feet are the main floor and the loft together? Square feet:	**360** (also accepted: 360 square feet, 360 square foot, 360 sq ft)	Loft: 2/7 × 280 = 80 square feet. Together: 280 + 80 = 360 square feet, still under 400.

PERSUASIVE WRITING AND ARGUMENTS (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
h1_s0	Set 1: *From Valentina's letter to the Library Board (a made-up letter with made-up numbers) 1Dear Library Board, Please keep our library open until 8 o'clock on weeknights. First, many students need a quiet place to do homework. On three different evenings last month, I counted more than 20 students waiting outside at closing time. 2Some people worry that longer hours would cost too much. However, the volunteer club has offered to help at the front desk. Besides, everyone at my school wants later hours, so you should agree.* Which sentence is a BANDWAGON trap? [Please keep our library open until 8 o'clock on weeknights. / On three different evenings last month, I counted more than 20 students waiting outside at closing time. / Some people worry that longer hours would cost too much. / Besides, everyone at my school wants later hours, so you should agree.]	**Besides, everyone at my school wants later hours, so you should agree.**	"Everyone at my school wants later hours" is about popularity, not proof. That is the bandwagon trap. The other sentences are her claim, her evidence, and the counterargument.
h2_s0	Set 1: Type your answer: "Please keep our library open until 8 o'clock on weeknights" is the main thing Valentina wants. It is her ___. (one word) Word:	**claim** (also accepted: the claim)	The claim is the main thing a writer wants the reader to agree with and do. Her reasons and evidence support it.

ID	Question	Answer	Why (explain to student)
h1_s1	Set 2: *From Mr. Percival's speech to the Library Board (a made-up speech with made-up numbers) 1I agree that students need a quiet place to study. But I ask the board to open the library on Saturday mornings instead of staying open late. The budget sheet says evening hours would cost twice as much as Saturday hours. 2Supporters of evening hours say Saturday is too far from the school week. They have a point. However, students could plan their big projects for Saturday. Also, teenagers never follow quiet rules, so evenings would be noisy.* Which explanation of the problem with "teenagers never follow quiet rules" is correct? [It is an overgeneralization: "never" claims something about every teenager, with no evidence. / It is a bandwagon trap: it says to agree because many people do. / It is strong evidence from the budget sheet. / It is a fair counterargument.]	**It is an overgeneralization: "never" claims something about every teenager, with no evidence.**	The word never claims something about every teenager, and he gives no evidence. That is an overgeneralization. His budget point is much stronger.
h2_s1	Set 2: Type your answer: in paragraph 2, Mr. Percival admits the other side has a point, then answers it with "However." A writer's answer to the other side's point is called a ___. Word:	**rebuttal** (also accepted with "a" or "an" or in the plural)	He states the other side's point fairly ("They have a point") and then answers it with "However, ...". A writer's answer to the counterargument is the rebuttal.
h1_s2	Set 3: *A summary of the Library Board meeting (made up for this checkpoint) 1Valentina wants the library to stay open until 8 o'clock on weeknights. Mr. Percival wants it to open on Saturday mornings instead, because evening hours cost more. 2Both speakers agree that students need a quiet place to study. They disagree about when the extra hours should be.* Two steps: Step 1, what do the two speakers mainly disagree about? Step 2, which COMPROMISE gives each side part of what it wants? [Whether students need a place to study; close the library. / When the extra hours should be; stay open late on two weeknights and also open one Saturday morning each month. / When the extra hours should be; stay open until midnight every night. / Whether libraries are useful; let everyone at school vote.]	**When the extra hours should be; stay open late on two weeknights and also open one Saturday morning each month.**	They agree that students need study space; they disagree about WHEN. A compromise gives each side part of what it wants: some evening hours and some Saturday hours. The other choices misstate the disagreement or ignore one side.
h2_s2	Set 3: Type your answer: an agreement in which each side gives up part of what it wants is a ___. Word:	**compromise** (also accepted with "a" or "an" or in the plural)	In a compromise, each side gives up part of what it wants so that both can agree.

LETTER WRITING (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
h3_s0	Set 1: Fix this letter greeting (add the end punctuation): Dear Library Board	**Dear Library Board,** (also accepted: Dear Library Board:)	A letter greeting ends with a comma. In a formal business letter, a colon is also correct.
h3_s1	Set 2: Fix this sentence: The late hours could of helped many students.	**The late hours could have helped many students.**	"Could of" is a mistake for "could have," which sounds similar when spoken quickly.
h3_s2	Set 3: Fix this sentence: The library is in Flagstaff Arizona near the pine forest.	**The library is in Flagstaff, Arizona, near the pine forest.**	Put a comma between the city and the state, and another comma after the state because the sentence keeps going.

PUTTING SKILLS TOGETHER (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Two steps: evaluate $(0.6 \times 0.5) \times 10^3$. Value:	**300**	Parentheses first: 0.6 × 0.5 = 0.30 (6 tenths × 5 tenths = 30 hundredths). Then × 10^3 moves every digit 3 places to the left: 0.3 → 3 → 30 → 300.
x2_s0	Set 1: Which explanation is correct? Why is 3/4 × 2.8 less than 2.8? [3/4 is less than 1, so the product is only part of 2.8. / Multiplying always makes a number smaller. / 2.8 is a decimal, so any fraction makes it smaller. / 3/4 has a smaller numerator than denominator, so you subtract.]	**3/4 is less than 1, so the product is only part of 2.8.**	A factor less than 1 takes only part of a number. 3/4 is less than 1, so 3/4 × 2.8 is only part of 2.8 (it is 2.1).
x1_s1	Set 2: Two steps: write an expression for "multiply 4 by the sum of 1/2 and 1/4," then evaluate it. What is the value? Value:	**3**	The words mean 4 × (1/2 + 1/4). Parentheses first: 1/2 + 1/4 = 2/4 + 1/4 = 3/4. Then 4 × 3/4 = 12/4 = 3.
x2_s1	Set 2: Which explanation is correct for $4.5 \div 10^2 = 0.045$? [$10^2 = 10 \times 2 = 20$, so divide 4.5 by 20 / $10^2 = 100$, so each digit moves 2 places to the right: 4.5 → 0.45 → 0.045 / Each digit moves 2 places to the left / Add two zeros to the end: 4.500]	**$10^2 = 100$, so each digit moves 2 places to the right: 4.5 → 0.45 → 0.045**	$10^2 = 10 \times 10 = 100$. Dividing by 100 moves each digit 2 places to the right: 4.5 → 0.45 → 0.045. A placeholder zero holds the tenths place.
x1_s2	Set 3: Two steps: work from the inside out. Evaluate 2 × [1.5 + (0.4 × 0.5)]. Value:	**3.4**	Parentheses: 0.4 × 0.5 = 0.2. Brackets: 1.5 + 0.2 = 1.7. Last step: 2 × 1.7 = 3.4.
x2_s2	Set 3: Which explanation is correct? Without computing, how does 2/3 × (845 + 96) compare with 845 + 96? [It is 3 times as large. / It is the same, because the parentheses stay. / It is smaller: it is only 2/3 as large, because 2/3 is less than 1. / It is larger, because multiplying always makes numbers bigger.]	**It is smaller: it is only 2/3 as large, because 2/3 is less than 1.**	The parentheses keep 845 + 96 as one group, and the whole group is multiplied by 2/3. 2/3 is less than 1, so the result is only 2/3 as large. No adding needed!

BE THE CHECKER (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x3_s0	Set 1: Find the mistake: Gwendolyn says 1.5 × 0.4 = 6.0. Find her mistake, then type the correct product. Product:	**0.6**	Multiply 15 × 4 = 60. Each factor has one decimal place, so the product has two: 0.60 = 0.6. Estimate: 1.5 × 0.4 is less than 1.5, because 0.4 is less than 1, so 6.0 cannot be right.
x4_s0	Set 1: Find the mistake! Ines writes, "Homer's Iliad tells how Thetis dipped baby Achilles in the River Styx." Which correction is accurate? [Ines is right; the Iliad tells all about the heel. / Achilles was dipped in the river Pactolus, not the Styx. / The Iliad never mentions the weak heel; the River Styx detail comes from later tellings. / Achilles was a Roman king, not a Greek warrior.]	**The Iliad never mentions the weak heel; the River Styx detail comes from later tellings.**	Homer's Iliad tells of Achilles' anger, but it never mentions a weak heel. The dip in the River Styx comes from later tellings. An accurate retelling says which version it follows.
x3_s1	Set 2: Find the mistake: Theodora says a 2 1/2-foot by 1 1/2-foot shelf has an area of 2 1/4 square feet, because 2 × 1 = 2 and 1/2 × 1/2 = 1/4. Type the correct area as a mixed number. Square feet:	**3 3/4** (also accepted, with or without the unit word: 15/4, 3.75, 3 and 3/4, 3-3/4, 3 6/8)	2 1/2 = 5/2 and 1 1/2 = 3/2. 5/2 × 3/2 = 15/4 = 3 3/4. Theodora left out two partial products: 2 × 1/2 = 1 and 1/2 × 1 = 1/2.
x4_s1	Set 2: Find the mistake! Augustin says, "If a writer admits the other side has a point, the argument gets weaker." Which explanation is correct? [Augustin is right; a writer should never mention the other side. / Admitting a point is a bandwagon trap. / A rebuttal means agreeing with everything the other side says. / Stating the counterargument fairly and then giving a rebuttal makes a writer more trustworthy, so the argument gets stronger.]	**Stating the counterargument fairly and then giving a rebuttal makes a writer more trustworthy, so the argument gets stronger.**	Stating the other side's point fairly, then answering it with a rebuttal, shows listeners you are fair, so they trust you more. It makes an argument stronger, not weaker.
x3_s2	Set 3: Find the mistake: Rosamund says 35 ÷ 10^3 = 0.35. Find her mistake, then type the correct quotient. Quotient:	**0.035**	10^3 = 1,000, so each digit moves 3 places to the right: 35 → 3.5 → 0.35 → 0.035. Rosamund moved only 2 places, which is dividing by 10^2.
x4_s2	Set 3: Find the mistake! Perpetua says, "A mystery told by a narrator outside the story, who calls everyone he or she, is told in the first person." Which explanation is correct? [That narrator tells the story in the third person; a first-person story is told by a character who uses I. / Perpetua is right; he and she are first-person words. / Mysteries never have narrators. / It is first person only if the detective is a girl.]	**That narrator tells the story in the third person; a first-person story is told by a character who uses I.**	A narrator who is not in the story and uses he, she, and names tells it in the third person. A first-person story is told by a character who uses I.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix this sentence: Dr. Fennimore asked, "where is the Moon tonight"?	**Dr. Fennimore asked, "Where is the Moon tonight?"**	The first word of a quoted sentence is capitalized, so Where needs a capital W. The question mark belongs to his words, so it goes inside the closing quotation mark.

ID	Question	Answer	Why (explain to student)
s2_s0	Set 1: Fix this sentence: The judges added there scores at the end.	**The judges added their scores at the end.**	Their shows ownership: the scores belong to the judges. There tells where something is.
s3_s0	Set 1: Fix this sentence (put the story's title in quotation marks): Our class read the story The Banana Mystery today.	**Our class read the story "The Banana Mystery" today.**	The title of a story goes inside quotation marks.
s4_s0	Set 1: Fix this sentence: Each of the runners carry a water bottle.	**Each of the runners carries a water bottle.** (also accepted: Each of the runners carried a water bottle.)	The subject is Each, which means each ONE, so it is singular and needs carries, not carry. The words "of the runners" do not change the subject.
s1_s1	Set 2: Fix this sentence (Ifeoma's words are an exclamation): "look at the dome"! Ifeoma shouted.	**"Look at the dome!" Ifeoma shouted.**	The first word of the quotation is capitalized. The exclamation point belongs to Ifeoma's words, so it goes inside the closing quotation mark, before the speaker tag.
s2_s1	Set 2: Fix this sentence: Your going to love the star show.	**You're going to love the star show.**	You're is short for you are: you are going to love the star show. Your shows ownership, as in your ticket.
s3_s1	Set 2: Fix this sentence: Mr. Ulrich the carpenter measured the board twice.	**Mr. Ulrich, the carpenter, measured the board twice.**	The carpenter renames Mr. Ulrich in the middle of the sentence, so it gets a comma on each side.
s4_s1	Set 2: Fix this sentence: The posters on the wall is bright.	**The posters on the wall are bright.** (also accepted: The posters on the wall were bright.)	The subject is posters (plural), not wall, so use are, not is. The words "on the wall" do not change the subject.
s1_s2	Set 3: Fix this sentence: valentina said, please keep the library open	**Valentina said, "Please keep the library open."** (also accepted: Valentina said, "Please keep the library open!")	Capitalize the name Valentina, put quotation marks around her exact words, capitalize the first word of the quote, and put the end mark inside the closing quotation mark.
s2_s2	Set 3: Fix this sentence: The robot lost it's wheel on the last turn.	**The robot lost its wheel on the last turn.**	The wheel belongs to the robot, so use the possessive its, with no apostrophe. It's means it is.
s3_s2	Set 3: Fix this sentence: The race was in Boulder Colorado on May 9 2026.	**The race was in Boulder, Colorado, on May 9, 2026.**	Put commas between the city and the state and after the state. Put a comma between the day and the year.
s4_s2	Set 3: Fix this sentence: Saoirse and Emeka studies the monarchs every fall.	**Saoirse and Emeka study the monarchs every fall.** (also accepted: Saoirse and Emeka studied the monarchs every fall.)	Saoirse and Emeka is a compound subject (two people), so it needs the plural verb study, not studies.

WRITER'S NOTEBOOK: MAKE YOUR CASE (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: Which sentence uses a transition with a comma AND avoids every reasoning trap? [However the planetarium is closer to our train stop. / However, everybody knows the planetarium is the best. / However, the planetarium is closer to our train stop. / However, only boring people like robots.]	**However, the planetarium is closer to our train stop.**	It has a comma after the transition However, and it gives a real reason. One choice is missing the comma, one is a bandwagon trap, and one attacks people instead of giving a reason.
k1_s1	Set 2: Which sentence punctuates a quotation correctly AND gives evidence? [Our guide said "Light travels about 186,000 miles in one second". / Our guide said, "Space is cool." / Our guide said, light travels about 186,000 miles in one second. / Our guide said, "Light travels about 186,000 miles in one second."]	**Our guide said, "Light travels about 186,000 miles in one second."**	A comma follows the speaker tag, the exact words are inside quotation marks, the period is inside, and the words give a fact from the Week 27 lesson. The others misplace punctuation, give no evidence, or leave out the quotation marks.
k1_s2	Set 3: Which sentence states a CLAIM and gives a real reason with because? [Our family should visit Boulder first because we could run on real cross-country trails. / Boulder is a city in Colorado. / Our family should visit Boulder first because everybody goes there. / Our family should visit Boulder first.]	**Our family should visit Boulder first because we could run on real cross-country trails.**	It states a claim (our family should visit Boulder first) and backs it with a real reason after because. One choice is only a fact, one uses a bandwagon reason, and one gives no reason.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match numerical expression	**numbers and operation symbols that name one value**	2 points if matched on the first try.
match_s0_2	Set 1: Match exponent	**the raised number that tells how many times the base is a factor**	2 points if matched on the first try.
match_s0_3	Set 1: Match allusion	**a quick reference to a well-known story**	2 points if matched on the first try.
match_s0_4	Set 1: Match red herring	**a clue that points readers the wrong way**	2 points if matched on the first try.
match_s0_5	Set 1: Match rebuttal	**a writer's answer to the other side's point**	2 points if matched on the first try.
match_s1_1	Set 2: Match order of operations	**the agreed rules for which step comes first**	2 points if matched on the first try.
match_s1_2	Set 2: Match placeholder zero	**a zero that holds an empty place**	2 points if matched on the first try.
match_s1_3	Set 2: Match synthesize	**to combine ideas from two or more sources**	2 points if matched on the first try.
match_s1_4	Set 2: Match alibi	**proof that a suspect could not have done it**	2 points if matched on the first try.
match_s1_5	Set 2: Match bandwagon	**a trap that says to agree because many people do**	2 points if matched on the first try.
match_s2_1	Set 3: Match divisor	**the number you divide by**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s2_2	Set 3: Match scaling	**resizing a number by multiplying it by a factor**	2 points if matched on the first try.
match_s2_3	Set 3: Match myth	**a traditional story that explains the world or teaches a lesson**	2 points if matched on the first try.
match_s2_4	Set 3: Match cite	**to name the source where you found a fact**	2 points if matched on the first try.
match_s2_5	Set 3: Match counterargument	**the other side's point, stated fairly**	2 points if matched on the first try.

UNIT 6 BOSS CHALLENGE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
z1_s0	Set 1: *The Situation Orsolya's robotics team in Detroit orders parts for next season: 4 wheels that cost $12.50 each and 10^2 bolts that cost $0.15 each. The 5 team members split the total cost equally. Then Orsolya writes a letter asking the city to build a robotics lab.* Three steps: how many dollars does each team member pay? Dollars:	**13** (also accepted: 13.00, 13 dollars, 13 dollar, 13.00 dollars, 13.00 dollar)	The expression is $(4 \times 12.50 + 10^2 \times 0.15) \div 5$. Wheels: $4 \times 12.50 = \$50$. Bolts: $10^2 = 100$, and $100 \times 0.15 = \$15$. Total: $50 + 15 = \$65$. Each member: $65 \div 5 = \$13$.
z2_s0	Set 1: Orsolya's draft says: Please build a robotics lab at the library. Last spring, 31 students joined the robotics club waitlist (club records). Everybody wants it, so you should say yes. Which revision keeps her evidence AND removes the reasoning trap? [Please build a robotics lab, because everybody wants one. / Please build a robotics lab at the library. / Please build a robotics lab at the library, because 31 students joined the robotics club waitlist last spring (club records). / Anyone who says no to a robotics lab does not care about students.]	**Please build a robotics lab at the library, because 31 students joined the robotics club waitlist last spring (club records).**	The best revision keeps the claim and the evidence with its source (the club records) and drops "Everybody wants it," which is a bandwagon trap. One choice keeps the trap, one drops the evidence, and one attacks people.
z1_s1	Set 2: *The Situation In Austin, Augustin builds a garden bed beside a tiny house. The bed is 4 1/2 feet long and 2 2/3 feet wide. He plants tomatoes in 3/4 of the bed, and each square foot of tomatoes needs 0.5 gallon of water a day. That night he writes a short mystery story.* Three steps: how many gallons of water do the tomatoes need each day? Gallons:	**4.5** (also accepted: 4 1/2, 9/2, 4 and 1/2, 4.5 gallons, 4.5 gallon, 4 1/2 gallons, 4 1/2 gallon, 9/2 gallons, 9/2 gallon, 4 and 1/2 gallons, 4 and 1/2 gallon)	Step 1: bed area: $9/2 \times 8/3 = 72/6 = 12$ square feet. Step 2: tomatoes: $3/4 \times 12 = 9$ square feet. Step 3: water: $9 \times 0.5 = 4.5$ gallons a day.

ID	Question	Answer	Why (explain to student)
z2_s1	Set 2: Augustin's mystery says: The muddy footprints by the door were the loudest clue, but they belonged to the mail carrier. "Obvious clues are my Achilles' heel," the detective sighed. Which answer explains the clue AND the allusion correctly? [The footprints are the real clue; "Achilles' heel" means the detective's heel hurts. / The footprints are a red herring; "Achilles' heel" means everything the detective touches succeeds. / The footprints are an alibi; "Achilles' heel" means the detective opened Pandora's box. / The footprints are a red herring; "Achilles' heel" means trusting obvious clues is the detective's one weak spot.]	**The footprints are a red herring; "Achilles' heel" means trusting obvious clues is the detective's one weak spot.**	The muddy footprints point toward the wrong person, so they are a red herring. "Achilles' heel" is an allusion: one weak spot in someone strong. The detective's weak spot is trusting the obvious clue.
z1_s2	Set 3: *The Situation At the Adler Planetarium in Chicago, a wall model divides real distances in miles by 10^6 and shows them in centimeters. Earth is about 93,000,000 miles from the Sun. For a smaller hallway model, Ifeoma makes every wall-model distance 2/3 as long. Then she writes a report about the planetarium.* Three steps: how many centimeters apart are Earth and the Sun in the hallway model? Centimeters:	**62** (also accepted: 62 centimeters, 62 centimeter, 62 cm)	Step 1: 10^6 = 1,000,000. Step 2: 93,000,000 ÷ 1,000,000 = 93 centimeters on the wall model. Step 3: 2/3 × 93 = 62 centimeters in the hallway model.
z2_s2	Set 3: Ifeoma has two sources. The planetarium sign says: The Adler Planetarium opened in Chicago in 1930. A guidebook says: It was the first planetarium in the Western Hemisphere. Which sentence SYNTHESIZES both and cites them correctly? [The Adler Planetarium opened in Chicago in 1930 (planetarium sign), and it was the first planetarium in the Western Hemisphere (guidebook). / The Adler Planetarium opened in Chicago in 1930 (guidebook), and it was the first planetarium in the Western Hemisphere (planetarium sign). / The Adler Planetarium opened in Chicago in 1930 (planetarium sign). / Everybody says the Adler is the best planetarium in the world (guidebook).]	**The Adler Planetarium opened in Chicago in 1930 (planetarium sign), and it was the first planetarium in the Western Hemisphere (guidebook).**	It combines a fact from each source in one sentence and cites each one correctly. One choice swaps the citations, one uses only one source, and one is a bandwagon claim, not a fact from either source.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade4-5/prepost/KCE_Checkpoint_PrePost_G4-5W30.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Evaluate 6 + 4 × (9 − 2); write "subtract 3 from 20, then multiply by 5"	6 + 4 × 7 = 6 + 28 = 34. Expression: (20 − 3) × 5.
2	10^4 in standard form; a) 3.7×10^2 b) $48 \div 10^3$	10,000. a) 370 b) 0.048

#	Prompt	Model answer / grading note
3	0.7 × 0.6; 4.8 ÷ 0.6	0.42; 48 ÷ 6 = 8
4	Area of a 3/4-foot by 2/3-foot shelf; is 5/4 × 16 greater or less than 16?	3/4 × 2/3 = 6/12 = 1/2 square foot. Greater, because 5/4 is greater than 1 (it is 20).
5	“Fixing my bike was a Herculean job.”	The myth of Hercules (Heracles) and his twelve almost impossible labors. It means fixing the bike was a huge, exhausting job.
6	Which sentence is a bandwagon trap?	Everyone in my class wants a garden, so build one.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	6 water bottles at $2.75 and 10^2 stickers at $0.05	6 × $2.75 = $16.50 and 100 × $0.05 = $5.00. Total: $21.50.
2	Elspeth says 3 1/2 × 1 1/2 = 3 1/4	She found only two of the four partial products and left out 3 × 1/2 = 1 1/2 and 1/2 × 1 = 1/2. Correct: 3 + 1 1/2 + 1/2 + 1/4 = 5 1/4 (or 7/2 × 3/2 = 21/4 = 5 1/4).
3	Be the editor	a) Dear Library Board, (a colon is also correct) b) The clues in the story point to the wind. c) Valentina said, "The library should stay open late."
4	Writing: convince your family which Unit 6 place to visit first (4–5 sentences)	Grading note: a clear claim naming a Unit 6 place (for example, Detroit, the Adler Planetarium in Chicago, Boulder, or Austin); a transition followed by a comma; a fact from the lessons as evidence; a counterargument answered with a rebuttal; no bandwagon, overgeneralization or attacking the person; 4–5 complete sentences.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	A model divides miles by 10^4; the Moon is about 239,000 miles from Earth; then a model half as long	239,000 ÷ 10^4 = 23.9 centimeters. Half as long: 23.9 ÷ 2 = 11.95 centimeters.